Our dataset is a CSV file containing 6,904 daily observations of historical weather data for Charlottesville, Virginia, from 2007–2026, obtained from the NOAA NCEI database. Each row represents one day of weather measurements, including maximum temperature (int), minimum temperature (int), and precipitation levels (float). For this project, the primary outcome variable is daily precipitation (PRCP) measured in inches. Monthly precipitation totals will be created by aggregating the daily precipitation values to produce a continuous monthly time series used for modeling.

Column	**Description**	**Potential Responses**
Date	The date	2/9/2026
TMAX (Degrees Fahrenheit)	The maximum temperature in degrees Fahrenheit on the given date	80
TMIN (Degrees Fahrenheit)	The minimum temperature in degrees Fahrenheit on the given date	44
PRCP (Inches)	The total precipitation in Charlottesville on the given day	1.57

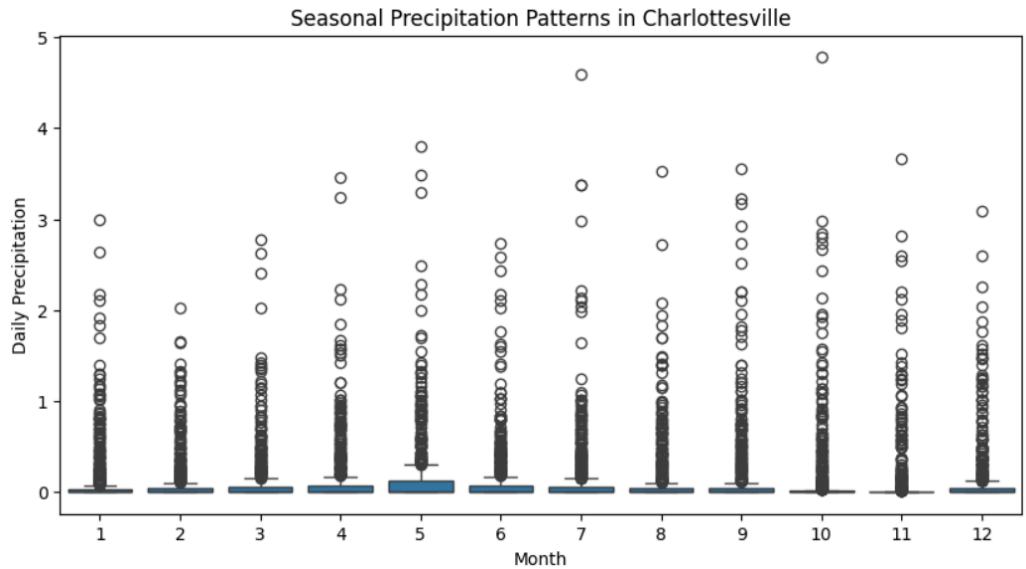
Summary Statistics:

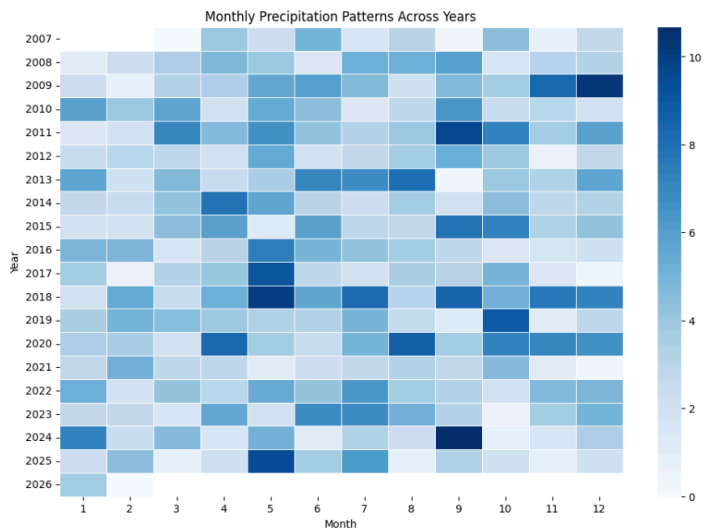
	TMAX (Degrees Fahrenheit)	TMIN (Degrees Fahrenheit)	PRCP (Inches)
count	6874.000000	6873.000000	6875.000000
mean	65.160023	48.916776	0.128356
std	16.522100	16.094790	0.368412
min	13.000000	-1.000000	0.000000
25%	53.000000	36.000000	0.000000
50%	67.000000	50.000000	0.000000
75%	79.000000	63.000000	0.040000
max	98.000000	80.000000	4.780000

Precipitation Frequency by Month:

```
month
1     64.57
2     54.60
3     63.50
4     76.12
5     96.46
6     76.22
7     80.72
8     71.96
9     85.03
10    76.79
11    59.60
12    75.20
Name: PRCP (Inches), dtype: float64
Highest total rainfall month: 5
Lowest total rainfall month: 2
```

Other plots:


Seasonal Precipitation Patterns in Charlottesville
Daily Precipitation
Month
0
1
2
3
4
5
1
2
3
4
5
6
7
8
9
10
11
12


Monthly Precipitation Patterns Across Years
Year
Month
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
1
2
3
4
5
6
7
8
9
10
11
12
0
2
4
6
8
10